

PR 1 — Foundations: what the three experiments (E1–E3) test, and why

Bayesian-Reflex implementation (branch `pr/foundations-e1-e3`)

Scope in one sentence. PR 1 validates the *scalar Gaussian inference module* of the Bayesian-reflex implementation: it checks that a single unknown number can be tracked with correct uncertainty (E1), updated one observation at a time without losing information (E2a), and that the prediction errors it produces are correctly scaled by that uncertainty (E3).

Not covered. The reflex loop, function-space inference, drift, action, and the paper’s look-up-table and ellipsoidal machinery. Those belong to later pull requests.

1 Why start here

The Bayesian Reflex (arXiv 2605.02825 and 2608.00492) rests on three ideas: *maintain a belief*, *update it sequentially*, and *act on its uncertainty*. Before testing any advanced machinery, the smallest possible version must be provably correct, because every later experiment is compared against an *oracle*, an exact reference answer. If the smallest version is wrong, nothing built on it can be trusted.

The scalar Gaussian case is ideal for this because the exact answer is known in closed form. E1–E3 are therefore closer to **rigorous unit tests of the implementation** than to research claims, and the pull request states this explicitly.

2 The model and the code under test

2.1 Model

One unknown number θ with a Gaussian prior, observed through Gaussian noise:

$$\theta \sim \mathcal{N}(\mu_0, s_0^2), \quad \mu_0 = 0, s_0 = 1 \text{ in E1 and E3}, \quad (1)$$

$$y_t = \theta + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma_y^2), \quad \sigma_y = 0.5 \text{ in E1; } \sigma_y \in \{0.1, 1.0\} \text{ in E3}. \quad (2)$$

2.2 Exact (conjugate) update in precision form

After each observation y_t , with posterior precision $\Lambda_t = 1/\sigma_t^2$:

$$\Lambda_t = \Lambda_{t-1} + \frac{1}{\sigma_y^2}, \quad (3)$$

$$\mu_t = \frac{\Lambda_{t-1} \mu_{t-1} + y_t / \sigma_y^2}{\Lambda_t}, \quad (4)$$

$$\sigma_t^2 = \frac{1}{\Lambda_t}. \quad (5)$$

2.3 Code components

All components live in `src/bayesian_reflex/`.

Component	Role
<code>GenerativeModel</code> (interface)	<code>update(observation)</code> , <code>predict()</code> , <code>sample(n, rng)</code> : the contract that every later belief model (GP, PC-FSVI, ...) implements.
<code>ScalarGaussianBelief</code>	The update above, vectorised over many independent trials at once.
<code>PredictiveCodingLayer</code>	<i>Reports</i> discrepancies: raw error $y - \hat{y}$, predictive precision, precision-weighted error, and the standardised error z .
<code>BayesianReflex</code>	Orchestrates one step: predict $\rightarrow$ compute error $\rightarrow$ update belief $\rightarrow$ (optionally) act.

Design point. The predictive-coding layer here only *reports* errors; it does not perform the Bayesian update. Whether the error should *drive* learning is an explicit open design decision, handled in a later experiment (E7).

3 E1 — Is the belief accurate and calibrated?

3.1 Question

Given noisy observations, does the posterior have the right mean *and* the right width? A belief can be accurate but over-confident. For a Bayesian system, uncertainty is useful only if it is calibrated.

3.2 Method: simulation-based calibration (SBC)

Checking one fixed θ would not test calibration: a fixed truth that falls in the tail of the prior makes intervals under-cover even with correct code. Instead:

1. draw the true $\theta^{(i)} \sim \mathcal{N}(\mu_0, s_0^2)$ for 1000 trials ($\times 10$ seeds = 10,000 trials);
2. generate $T = 50$ observations for each trial from that $\theta^{(i)}$;
3. update the belief *sequentially*, one observation at a time;
4. at the checkpoints $t \in \{1, 5, 10, 25, 50\}$, ask where the true $\theta^{(i)}$ falls in the posterior.

If the implementation is correct, the true θ behaves like one more draw from the posterior, so its *rank* among L posterior samples is uniform:

$$r_i = \#\{\ell \leq L : \theta_\ell^{(i)} < \theta^{(i)}\} \sim \text{Uniform}\{0, \dots, L\}. \quad (6)$$

Here $L = 99$ posterior draws per trial are used and ranks are binned into 10 bins.

3.3 Metrics and pre-fixed criteria

Check	Criterion
50% / 80% / 95% interval coverage	within 3 binomial standard errors of nominal (tolerances ± 0.015 / ± 0.012 / ± 0.0065 at 10,000 trials)
Rank uniformity	χ^2 test on the 10-bin rank histogram, $p > 0.001$
Probability-integral transform (PIT)	Kolmogorov–Smirnov test against Uniform(0, 1), $p > 0.001$
Accuracy against claimed uncertainty	$\text{RMSE}(\mu_t) / \sqrt{\sigma_t^2}$ within 3% of 1 (for a correct posterior $\mathbb{E}[(\mu_t - \theta)^2] = \mathbb{E}[\sigma_t^2]$)
Closed-form sanity	the code’s posterior standard deviation equals $(1/s_0^2 + t/\sigma_y^2)^{-1/2}$

3.4 Negative control (must fail)

The same belief, told the wrong noise level (σ_y assumed 0.25 while the data use 0.5). If the test cannot detect this, it proves nothing.

3.5 Secondary result (not pass/fail): prior mismatch

With true $\theta = 2.0$, a prior centred at 10 needs 21 observations before the mean bias falls below 0.1, against 5 for a prior centred at 0. Its 95% coverage at $t = 1$ is 0.03. This shows how long a bad prior keeps its influence.

3.6 Result

32 of 32 checks pass. The 95% coverage is 0.950–0.953 at every checkpoint, and RMSE/sd lies in 0.985–1.009. For the control, the 95% coverage collapses to 0.66 and the KS test rejects ($p \approx 0$).

3.7 Interpretation and limits

For a conjugate Gaussian model, calibration is guaranteed analytically, so passing shows the *code* is right, not that nature is Gaussian. The same SBC harness is reused later for non-Gaussian methods.

4 E2a — Does sequential updating equal batch updating?

4.1 Question

The central property of recursive Bayes (paper Theorem 2.1): processing observations one by one must give exactly the posterior obtained from all data at once,

$$p(\theta \mid y_{1:t}) \propto p(y_t \mid \theta) p(\theta \mid y_{1:t-1}). \quad (7)$$

Any information loss or numerical drift would show up here.

4.2 Method

For 100 random problems (random prior mean, prior standard deviation, noise level, and true θ), run the sequential update for $T = 100$ observations and compare at *every* step with a **batch oracle**. The oracle is the closed-form posterior of the first t observations,

$$\Lambda_t^{\text{batch}} = \frac{1}{s_0^2} + \frac{t}{\sigma_y^2}, \quad \mu_t^{\text{batch}} = \frac{\mu_0/s_0^2 + \sum_{k \leq t} y_k/\sigma_y^2}{\Lambda_t^{\text{batch}}}, \quad (8)$$

computed in extended precision (`longdouble`), so it is independent of the double-precision recursion being tested.

4.3 Metrics

The absolute difference in mean, the absolute difference in variance, and

$$\text{KL}(\mathcal{N}(\mu^{\text{seq}}, \sigma^{2,\text{seq}}) \parallel \mathcal{N}(\mu^{\text{batch}}, \sigma^{2,\text{batch}})), \quad (9)$$

evaluated in a cancellation-safe form (using `log1p`).

4.4 Pre-fixed criteria and observed values

Check	Criterion	Observed
Max difference in mean	$< 10^{-6}$	1.7×10^{-14}
Max difference in variance	$< 10^{-6}$	3.2×10^{-16}
Max KL	$< 10^{-12}$	7.9×10^{-26}
Order independence (shuffled data)	$< 10^{-10}$	4.4×10^{-15}
Negative control: drop every 10th observation	must differ ($> 10^{-3}$)	2.9×10^{-2}
Stress test $T = 100,000$ (20 problems): mean / variance / KL	finite, same criteria	9.4×10^{-12} / 2.5×10^{-12} / 1.6×10^{-17}

4.5 Result

9 of 9 checks pass.

4.6 Honest note

The mean error grows roughly linearly with the number of steps (log-log slope ≈ 1.06), the expected signature of accumulated rounding. It stays far below 10^{-6} even at 100,000 steps, but this does not prove stability for arbitrary horizons.

5 E3 — Are prediction errors correctly scaled by the belief’s uncertainty?

5.1 Question

E3 is the first predictive-coding quantity. Before seeing y_t , the belief predicts it with mean $\hat{y}_t = \mu_{t-1}$ and *predictive* variance $\sigma_{t-1}^2 + \sigma_y^2$, which is parameter uncertainty *plus* observation noise. The standardised error

$$z_t = \frac{y_t - \hat{y}_t}{\sqrt{\sigma_{t-1}^2 + \sigma_y^2}} \quad (10)$$

should then be exactly $\mathcal{N}(0, 1)$ and independent over time if the belief is correct. This formalises the statement that a 0.5 error means something different when the uncertainty is 0.1 than when it is 2.0.

Important detail. z_t uses the variance from *before* the update. Using the post-update variance is a classic mistake and is tested as a control.

5.2 Setup

θ drawn from the prior; 1000 trials $\times$ 10 seeds $\times$ $T = 50$; two noise levels, $\sigma_y \in \{0.1, 1.0\}$.

5.3 Checks (for each σ_y)

Check	Criterion
Mean of z	within 3 SE of 0
Variance of z	within 3 SE of 1
Lag-1 autocorrelation	within 3 SE of 0 (innovations must be independent)
Distribution	KS test against $\mathcal{N}(0, 1)$ at $t = 1, 5, 25, 50$, $p > 0.001$
Kalman identity	the belief update equals gain $\times$ raw error, with gain = $\frac{\sigma_{t-1}^2}{\sigma_{t-1}^2 + \sigma_y^2}$, to $< 10^{-12}$ (observed 9×10^{-16})

Explicitly, the identity being checked is

$$\mu_t = \mu_{t-1} + \frac{\sigma_{t-1}^2}{\sigma_{t-1}^2 + \sigma_y^2} (y_t - \mu_{t-1}). \quad (11)$$

5.4 Precision weighting made visible

The spread of the raw error follows σ_y (ratio 10.0 between $\sigma_y = 1.0$ and $\sigma_y = 0.1$), while the spread of z does not (ratio 1.000). The standardised error removes the noise scale, as intended.

5.5 Negative controls (must fail)

- *Wrong σ_y* ($0.5 \times$ the true value): $\text{Var}(z) \approx 3.9$ instead of 1.
- *Stale variance* (post-update instead of pre-update): $\text{Var}(z)$ over the first five steps is 11.0 for $\sigma_y = 0.1$ and 1.12 for $\sigma_y = 1.0$, and the KS test rejects at $t = 1$.

5.6 Extension (paper §5.2): unknown noise variance

With a Normal–Inverse-Gamma prior the predictive distribution is a **Student- t** . With hyperparameters (m, κ, a, b) after $t - 1$ observations,

$$y_t \mid y_{1:t-1} \sim t_{2a_{t-1}} \left(m_{t-1}, \frac{b_{t-1}(\kappa_{t-1} + 1)}{a_{t-1} \kappa_{t-1}} \right), \quad (12)$$

so z_t should follow a t distribution with $2a_{t-1}$ degrees of freedom, not a Gaussian. The PIT through the t CDF passes ($p = 0.06$ – 0.79); using the Gaussian CDF instead is rejected ($p \approx 6 \times 10^{-24}$), which is the control.

5.7 Result

29 of 29 checks pass.

5.8 Caveats

Two statistics at $\sigma_y = 0.1$ (the mean of z and the lag-1 autocorrelation) sat close to their limits. Rerunning $\sigma_y = 1.0$ on two fresh sets of 10 seeds put both well inside, so this looks like chance. The experiment does *not* show that the predictive-coding layer drives learning; it only shows that its error signal is correctly standardised.

6 What PR 1 claims, and what it does not

Claims (exactly these):

1. The scalar Gaussian belief is accurate and calibrated (E1).
2. Sequential updating reproduces the batch posterior to numerical precision, including a 100,000-step stress test (E2a).
3. One-step-ahead prediction errors are correctly precision-weighted under the Bayesian predictive, including the Student- t case for unknown variance (E3).

Does not claim: anything about function-space belief, drift, uncertainty-driven action, the closed loop, or the paper-specific look-up-table and ellipsoidal sampler.

Method rules (carried through all later pull requests):

- pass/fail criteria are fixed *before* running;
- every positive test has a *negative control that must fail*;
- at least 10 seeds, with tolerances derived from confidence intervals;
- claims are limited to what a stage actually tests.

7 Reproduce

```
pip install -r requirements.txt
pytest # 3 passed
python experiments/e1_scalar_belief.py # ends with "...foundation only): PASS"
python experiments/e2_sequential_equivalence.py # ends with "...vs batch oracle):
    PASS"
python experiments/e3_prediction_error.py # ends with "...prediction error): PASS"
```